

KICC National Exhibition Platform

Full Engineering & Achievements Report

Reporting Period: August 2 – August 20, 2026 | Report Date: August 21, 2026

EXECUTIVE SUMMARY

Over the past 18 days, the Graphics & R&D Department and the Platform Engineering team delivered a production-grade digital twin of the Kenyatta International Convention Centre and all 47 Kenyan counties. The platform spans a Laravel production site, a Kotlin 4-tier engine, Cloudflare Workers edge gateway, R2 media pipeline, marketplace commerce, trade system, AI-powered 3D/immersive content pipeline, and 47-county admin infrastructure.

SECTION 1: PLATFORM ENGINEERING — KICCTEST.ORG

1.1 Live Production Site

kicctest.org is live and serving traffic through a Cloudflare edge network with PHP 8.4, TiDB Cloud production database, and Laravel Cloud deployment. The site handles 47 county public pages, a national marketplace, trade system, and admin panels.

1.2 47-County System

All 47 Kenyan counties are fully onboarded with individual public websites, each containing: sector-specific content, tourism attractions, hotels, product listings, and trade data. Counties are organized by region (Central, Coast, Eastern, Nairobi, North Eastern, Nyanza, Rift Valley, Western) with a searchable, filterable directory.

1.3 Commerce & Marketplace

Full marketplace with product listings, variants, pricing, stock tracking, and category associations. Orders are processed through an escrow system with buyer/seller protection. Supports search, filtering, booking flows, and complete order management.

1.4 Trade System

Comprehensive trade module connecting Kenyan counties to international markets across trading blocs: EAC, COMESA, AfCFTA, and WTO. Export eligibility checker determines tariff preferences. Export enquiry workflow creates trade lead records integrated with n8n automation.

1.5 Admin Control Panels

KICC Mother Admin: central god-mode dashboard overseeing all 47 counties, national government exhibitors, private exhibitors, marketplace orders, provider certifications, escrow transactions, users, and landing page hero media. Includes Artisan command console, provider certification queue, and escrow release controls.

47 County Professional Admin Dashboards: each county manages content, images, prices, advertising, 4D immersive videos, sectors, entities, and CSV reports.

1.6 Media Asset Pipeline

Centralized MediaAsset system managing every image, video, and 3D model in Cloudflare R2 storage, served via CDN with CORS headers, range request support, and immutable cache. Supports: posters, thumbnails, WebP, HLS streaming, 4D immersive video slots.

1.7 Infrastructure & DevOps

Cloudflare Workers edge gateway (caching, rate limiting, JWT). Nginx + PHP-FPM with 1GB upload support. GitHub Actions CI/CD. Laravel Horizon, Pulse, Sentry. R2 media CDN.

1.8 Security & Authentication

Email verification, Google OAuth, MFA/TOTP, role-based access (kicc_admin, county_admin, exhibitor, provider), Sanctum tokens, Passport OAuth2, rate-limited edge, CSRF.

1.9 Testing & Quality

148 tests, 202 assertions. Database scalability indexes. Security penetration test suite. OpenAPI spec at /api/openapi.

SECTION 2: GRAPHICS & R&D — 4D VIDEO & 3D PIPELINE

2.1 AI Model Integration & API Aggregation

OpenRouter gateway for 200+ models. Anthropic Claude for code generation. DeepSeek for integration scripts. Moonshot AI (Kimi) for large-context processing.

2.2 Cloud GPU Compute & Distributed Rendering

Vast.ai GPU instances deployed via automated script (deploy_vast.py) for neural rendering, Gaussian splatting, and COLMAP SfM. GPUs with 16GB+ RAM, 50GB+ disk, under \$0.50/hr.

2.3 3D Reconstruction & Photogrammetry

Full pipeline: photogrammetry.py (SfM), splat_worker.py (Gaussian splatting), stereo_worker.py (depth estimation), depth_worker.py (depth maps), blender_3d_walkthrough.py (Blender rendering).

2.4 Multi-Agent Orchestration

Agentic loop: actor, decider, observer, learner, workflow_engine. N8n with 3 production workflows: county-content-pipeline, trade-promotion, sbs-pipeline.

2.5 2.5D/3D Content Generation

Automated pipeline: auto_pipeline.py, orchestrator.py, orchestrator_v2.py. Components: image_analyzer, refiner, interpolator, scene_composer, terrain_gen, video_service.

2.6 Model Holders

TRELLIS, TRELLIS.2 for 3D representation learning. img2threejs with PBR extraction. Three.js graphics agent skills.

2.7 Cinematic Hero Video

81-second, 7-scene walk-through: Main Gate push-in, reception, concourse, staircase, Tsavo Hall, Amphitheater montage, Lenana Room, finale. TV-news overlays, Ken Burns zooms, color grading. 21MB, CORS-enabled, streamed via CDN.

2.8 Workflow Automation

N8n Cloud: county content pipeline, trade promotion, stereoscopic processing.

SECTION 4: EXPENDITURE TABLE — DIGITAL SERVICES

Category	Item	Amount (KES)
AI Models & API Gateway	OpenRouter (AI API Gateway)	KES 121,330.06
	Anthropic / Claude (AI Models)	KES 6,360.22
	DeepSeek (AI Models)	KES 725.64
	Moonshot AI / Kimi (AI Models)	KES 670.00
Generative Video & 3D	Runway ML (Gen Video Platform)	KES 16,680.56
	Luma AI & API Credits (Dream Machine)	KES 19,205.13
	Depthflow / DP DodoPay (2.5D Tool)	KES 2,056.10
	DaVinci AI (Video Editing AI)	KES 272.11
Upscaling & Image	Magnific AI (Gen Upscaler)	KES 8,085.06
	Adobe Creative Cloud (Asset Tooling)	KES 5,389.57
Cloud GPU Compute	Vast.ai (GPU Marketplace)	KES 18,230.38
	RunPod.io (Cloud GPU)	KES 2,744.66
Cloud Infrastructure	TiDB Cloud / PingCap (Database)	KES 137.23
	Cloudflare (Edge / DNS / CDN)	KES 2,557.41

Creative Assets	Artlist.io (Music / SFX Library)	KES 11,097.62
Automation & Ops	N8n Cloud (Workflow Automation)	KES 3,811.70
	Telegram Premium (Bot Notifications)	KES 3,973.31
	Google One (Cloud Storage)	KES 3,700.00
TOTAL		KES 227,026.76

Note: These figures are drawn from the Graphics & R&D Department's certified bank statements. Items marked with an asterisk are third-party services whose invoices were processed during the reporting period. All amounts in Kenyan Shillings (KES).

SECTION 5: PHYSICAL EQUIPMENT PROCUREMENT

The following production and field equipment was procured for the Graphics & R&D Department's content capture, 3D photogrammetry, and video production operations.

#	Item Description	Qty	Invoice / Receipt #	Total (KES)
1	Canon EOS C50 Body (SN: 043119000360)	1	Invoice 5927 / DN 5872	465,000.00
2	Canon EOS R5 Mark II Body (SN: 990002102494)	1	Invoice 5927 / DN 5872	383,000.00
3	Canon RF 70-200mm f/2.8 L IS USM (SN: 58600037)	1	Invoice 5927 / DN 5872	264,000.00
4	Insta360 X5 BMW Motorrad Action Camera (SN: 1A)	1	Invoice 5927 / DN 5872	64,000.00
5	Canon RF 24-70mm f/2.8 L IS USM (SN: 664400189)	1	Invoice 5927 / DN 5872	238,000.00
6	Lexar 512GB CFexpress Type B 8K Silver Card	4	Invoice 5927 / DN 5872	112,000.00
7	Canon LP-E6P Original Battery	2	Invoice 5927 / DN 5872	36,000.00
8	Sirui SH25 Video Tripod	2	Invoice 5927 / DN 5872	40,000.00
9	K&F Concept KF13.210 Nature Wander 05 25L Back	1	Invoice 5927 / DN 5872	13,000.00
10	Lexar RW550 CFexpress Type B Card Reader	1	Invoice 5930 / DN 5874	8,000.00
11	SanDisk 4TB Extreme Portable SSD 1050MB/s	3	Invoice 5930 / DN 5874	225,000.00
12	Sony MDR-7506 Headphones	3	Invoice 5930 / DN 5874	48,000.00
13	Transcend 2TB StoreJet 25M3 External Hard Drive	2	Invoice 5930 / DN 5874	28,000.00
14	Godox V100 Flash	2	Invoice 5930 / DN 5874	70,000.00
15	Lexar 256GB MicroSD Card Silver Plus	2	Invoice 5930 / DN 5874	17,000.00
16	DJI RS 4 / RS 5 Combo Gimbal	1	Invoice 5930 / DN 5874	90,000.00
17	Souider 2022 Bag with Wheels	1	Invoice 5930 / DN 5874	12,000.00
18	LightWave USB-C Multifunction 8-in-1 Dock Station	2	Invoice 5930 / DN 5874	8,000.00
19	Dell Precision 5560 Core i7 11th Gen (16GB/512GB)	1	Receipt 3100 (Soke Computers)	80,000.00
20	Oraimo 40,000 mAh Power Bank	2	Cash Sale 728 (Mobi Arena)	8,000.00
21	Huawei 5G Indoor CPE H153-381 + UPS (SN: ...897)	1	Receipt 059083378 (Safaricom)	2,999.00
22	Huawei 5G Indoor CPE H153-381 + UPS (SN: ...275)	1	Receipt 059098339 (Safaricom)	2,999.00

COMBINED GRAND TOTAL (Services + Equipment): KES 2,442,024.76

All equipment procured under Graphics & R&D Department budget. Items verified against delivery notes and serial numbers.

SECTION 6: PROPOSED MONTHLY SUBSCRIPTION BUDGET — PRODUCTION SCALE-

FORECAST / PROPOSED — NOT YET ACQUIRED

The following subscriptions are required to scale up graphics and video production from the trained models. These are projected monthly costs (unless otherwise noted) for the next production phase.

Magnific AI (Gen Upscaler)	\$450	Monthly
Luma AI (Dream Machine / Spatial AI)	\$200	Monthly
KIRI Engine (3D Scanner / Photogrammetry)	\$1,000	Rollover
Marble (3D Reconstruction)	\$100	Monthly
Runway (Gen Video Platform)	\$120	Monthly
Vast.ai GPU Instance	\$3,500	Monthly
Seedance 2.0 / Veo 3.1 (Video Gen)	\$200	Limited credits / pay-as-you-go
OpenRouter (AI API Gateway)	\$800	Monthly

Topaz AI (Video Upscaler)	\$200	Monthly
Tripo AI (3D Model API)	\$1,000	API per call (\$0.40) / avg monthly
Rollover Cost Allowance	\$200	—
Artist (Music / SFX Library)	\$400	—
Firefly (Adobe Gen AI)	\$300	—

Subscription services for generative video production, 3D reconstruction, GPU compute, and AI model access. All amounts in USD as invoiced by respective providers.

ADDITIONAL EXPENDITURE: MURANGA COUNTY REMUNERATIONS

Muranga County Data Collection & Field Operations Remuneration	KES 190,000.00
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ADDITIONAL EQUIPMENT PROCUREMENT — DATA COLLECTION & PRODUCTION

Supplementary equipment for expanded data collection capacity and high-performance graphics workstation for the assistant software developer.

1. Electronics & Computing			
1	Apple MacBook Pro (M5, 32GB RAM, 1TB SSD)	1	375,000.00
2	Ugreen CM681 11-in-1 USB-C Dock	1	11,500.00
Section Subtotal			386,500.00
2. Cameras, Lenses & Video Gear			
3	Canon EOS R6 Mark III (Body Only)	1	340,000.00
4	Canon RF 15-35mm f/2.8 L IS USM	1	300,000.00
5	Canon RF 100-300mm f/2.8 L IS USM	1	1,350,000.00
6	DJI Mavic 4 Pro (RC2) Combo	1	265,000.00
7	DJI Mavic 4 Extra Intelligent Flight Battery	1	24,000.00
8	Canon LP-E6NH Batteries (USB-charging)	4	54,000.00
9	Canon BG-R20 Battery Grips	3	150,000.00
Section Subtotal			2,483,000.00
3. Audio Equipment			
10	Rode NTG5 Shotgun Microphone Kit	1	50,000.00
11	Hollyland Lark Max 2 Combo (4-Person)	1	45,000.00
Section Subtotal			95,000.00
4. Storage Media			
12	Lexar 512GB Silver Plus microSD	2	20,000.00
13	Lexar CFexpress Type B 512GB Silver	2	44,000.00
Section Subtotal			64,000.00
5. Lighting & Studio Support			
14	Nanlite FC-720B Bi-color LED Spotlight	2	500,000.00
15	Visico LS-8010 Heavy Duty Boom Stand	2	56,000.00
16	Phottix Raja 150cm Softbox	2	84,000.00
17	Kingjoy MP3008 Multifunctional Monopod	1	15,000.00
18	Insta360 Invisible Selfie Stick + Tripod	1	7,500.00
Section Subtotal			662,500.00
6. Power & Accessories			
19	EcoFlow Delta 2 Max Portable Power Station	1	212,500.00
20	Premium Leather Camera Straps	2	13,000.00
Section Subtotal			225,500.00
7. Software & Post-Production			
21	DaVinci Resolve Studio (Perpetual License)	1	50,000.00
22	Imagen AI & Colourlab AI (Est. Monthly)	—	8,000.00
Section Subtotal			58,000.00
Additional Equipment Subtotal			KES 3,974,500.00
Muranga Remunerations Subtotal			KES 190,000.00

COMBINED GRAND TOTAL (All Spots) KES 6,606,524.76

Previous total (Section 4 + 5): KES 2,442,024.76 | Additional equipment: KES 3,974,500.00 | Muranga remunerations: KES 190,000.00 | Proposed subscriptions: USD 8,470/month

FUNDING REQUEST SUMMARY — NEXT PHASE

Item	Amount
Additional Equipment (Data Collection & Production)	KES 3,974,500.00
Muranga County Remunerations	KES 190,000.00
Monthly Subscriptions (Magnific, Luma, KIRI, Marble, Runway, Vast, Seedance, Open	USD 8,470/mo
TOTAL ADDITIONAL FUNDING REQUIRED	KES 4,164,500.00 + USD 8,470/mo

SECTION 7: INTEGRATION ROADMAP — APIs READY TO CONNECT

The following services have full integration pipelines built but require API keys or configuration to go live. Each has a dedicated service class, controller wiring, and graceful fallback to stub mode.

1	KRA PIN Verification	Auto-verify business registration for exhibitors	Stub built
2	Huduma Namba / eCitizen	Verify national ID during registration	Stub built
3	NTSA	Verify vehicle registration & driving licenses	Stub built
4	Kenya Wildlife Service	Real-time park data, fees, conservation status	Stub built
5	Ministry of Tourism	Official star ratings for hotels	Stub built
6	National Museums of Kenya	Heritage site data per county	Stub built
7	Ministry of Health	Verify health facility registrations	Stub built
8	Ministry of Agriculture	Real-time crop prices & market data	Stub built
9	Ministry of Education	Verify institution accreditation	Stub built
10	TIMBR (Business Registry)	Auto-verify business names	Not started
11	NEMIS (Education MIS)	School enrollment data per county	Not started
12	iTax (KRA)	Tax compliance for marketplace sellers	Not started
13	M-Pesa STK Push	Accept mobile money payments	Stub mode
14	Stripe	International card payments	Stub mode
15	Airtel Money	Alternative mobile money	Stub mode
16	T-Kash	Telkom Kenya mobile money	Stub mode
17	Crypto (Onramp)	Cryptocurrency payments	Stub mode
18	OpenWeather	Live weather data per county	API key missing
19	Elasticsearch	Full-text search with typo tolerance	Fallback to MySQL
20	Redis	Production caching & queue	Disabled
21	Africa's Talking SMS	Transactional SMS notifications	Sandbox only
22	Africa's Talking USSD	Feature-phone menu for counties	Not deployed
23	Wan2GP Engine	Video-to-video generation pipeline	Disabled
24	Hailuo Engine	AI video generation from text/images	Disabled
25	Kling Engine	Alternative video generation	Disabled
26	Tripo3D Engine	3D model generation from single image	Disabled
27	ROAD Engine	Neural 3D scene reconstruction	Disabled

27 integrations identified. 9 government APIs + 6 payment drivers + 6 pipeline engines + 3 comms + 3 infra

SECTION 8: N8N WORKFLOW AUTOMATIONS

Three production automation workflows built in N8N format, ready for deployment once N8N server is configured. Additionally, 26 platform events are already wired to fire webhooks via N8nService.

County Content Pipeline	Webhook (on content update)	Auto-validate images, process with AI, store in R2, push to international fe
International Trade Promotion	Scheduled (weekly)	Fetch export-ready products, generate promo via OpenAI, publish to trade
SBS 3D Pipeline	Webhook (on 3D upload)	Scene analysis via OpenRouter, queue depth mapping, generate SBS 3D v

26 platform events already wired for N8N webhooks: user_registered, order_created, order_paid, escrow_released, product_listed, trade_enquiry_created, ky

SECTION 9: SEO OPTIMIZATION STATUS

1	Meta titles & descriptions	Done	Basic per-page setup
2	Open Graph tags	Done	og:title, og:description, og:image
3	Canonical URLs	Done	Present on public pages
4	Sitemap.xml route	Done	Route exists
5	Hero video poster	Done	Poster loads before video
6	CDN cache headers	Done	Immutable cache on media
7	Cloudflare edge caching	Done	HTML cache with SWR
S1	Dynamic sitemap per county	HIGH	Need individual sitemaps for each county + sector pages
S2	Schema.org JSON-LD	HIGH	Organization, LocalBusiness, Product, TouristAttraction, Event
S3	SEO-friendly URLs	MED	Clean slugs, trailing slashes, no query params on public pages
S4	Hreflang tags	MED	en-KE primary, en-US/en-GB for international trade content
S5	Page speed optimization	HIGH	Lazy-load images, defer JS, inline critical CSS, optimize LCP
S6	AI image alt tags	MED	Auto-generate alt text for all county images using AI captions
S7	Breadcrumb navigation	MED	Structured breadcrumbs: County > Sector > Product
S8	Preconnect hints	LOW	DNS-prefetch for R2 CDN, fonts, OpenRouter
S9	Core Web Vitals	MED	Set up Lighthouse CI to track CLS, LCP, FID per deploy
S10	Google Business Profile sync	MED	Sync county data with Google Business Profiles for local SEO
S11	News sitemap	LOW	For KICC news/blog section

SECTION 10: KEY ACHIEVEMENTS SUMMARY

- Platform: Live production Laravel application serving kicctest.org through Cloudflare edge network.
- Counties